

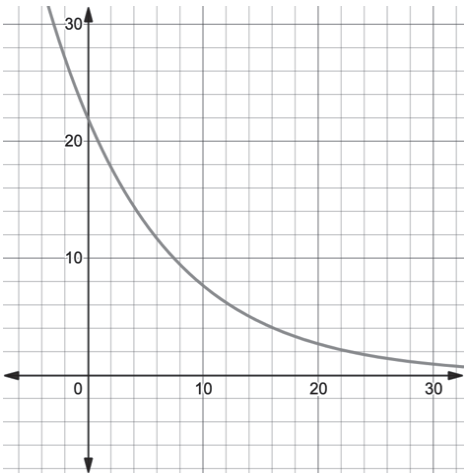
Algebra 1
Unit 11
Lesson 1 Practice

Name _____
Date _____
Period _____

1. In the equation $y = a(b)^x$, a represents the ☐ growth factor and b represents the ☐ initial amount.
2. When b is between 0 and 1, the equation models ☐ exponential growth. ☐ exponential decay.
3. When b is greater than 1, the equation models ☐ exponential growth. ☐ exponential decay.

For questions 4-8, identify if each representation of a function is an example of exponential growth or exponential decay. Then justify your selection.

4. $y = 142(1.05)^x$ represents _____ because...

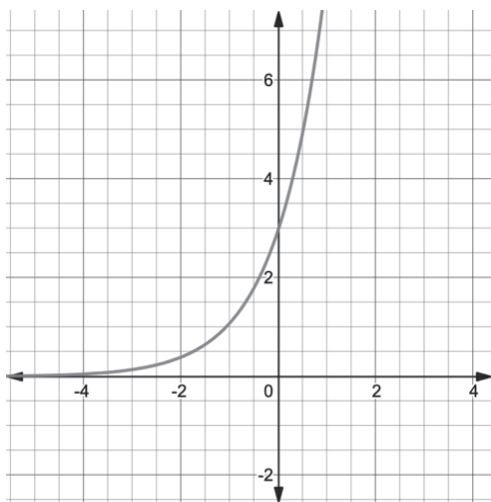
5.  represents _____ because...

6. _____ represents _____ because...

x	y
-1	$\frac{1}{5}$
0	$\frac{1}{2}$
1	$\frac{5}{4}$
2	$\frac{25}{8}$

7. $y = 3(1 - 0.14)^x$ represents _____ because...

8. _____ represents _____ because...



9. Write an equation in $y = a(b)^x$ form that represents exponential growth.

10. Write an equation in $y = a(b)^x$ form that represents exponential decay.